Muawiya Amir

Software Engineer · Full Stack & Open Source

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PROFESSIONAL SUMMARY

Software engineer with full-stack experience at SudoStudy and a strong open-source background. I built tools that help teachers create quizzes, grade work, and understand student progress. As a Triage Collaborator on drt-hub/drt, I build connectors and observability tools and review community contributions. I study Artificial Intelligence and Mathematics and enjoy making complex systems useful to people.

TECHNICAL SKILLS

Languages: Python, C / C++, JavaScript, TypeScript, Java, SQL, x86_64 Assembly

AI / ML: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Computer Vision, NLP

Web & Backend: React, Node.js, Flask, REST APIs, HTML / CSS

Systems & DevOps: Git, GitHub Actions, Docker, CI/CD, pytest, JSON Schema, OpenTelemetry

CS Foundations: Operating Systems, Data Structures, Algorithms, Linear Algebra, Probability

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack), SudoStudy — Sep 2025 — Oct 2026

- Built teacher workflows for section-based quizzes, student assignment, Google Classroom publishing, and grade return.
- Developed scoring and performance reports with charts and PDF exports.
- Improved a Gemini study assistant with streaming responses and conversation history.
- Improved mobile teacher dashboards, fixed authentication and grading issues, and contributed to read-only teacher analytics that protects student contact details.

Open-Source Engineer & Triage Collaborator, drt-hub/drt — Apr 2026 — Present

- Promoted to Triage Collaborator after sustained, high-quality contributions; review pull requests, triage issues, and shape feature design with maintainers.
- Designed and built the dynamic connector registry that auto-discovers sources and destinations, removing hardcoded dispatch maps across the engine.
- Added the Jira connector (create/update), JSON Schema validation for config files, and data-quality tests (freshness, unique, accepted_values).
- Shipped OpenTelemetry tracing and metrics with lazy, idempotent providers (+680 lines) and dry-run row-count diffs for safe sync previews.

PROJECTS

One Concept — An Android app that teaches one short technical concept each day. It rotates topics, sends timely reminders, and keeps a synced record of what you have learned.

Diskern — A desktop app that explains what is safe to clean before moving files into a recoverable quarantine. Built with a Rust rules engine and a Tauri/React interface.

BSAI University Projects (NFC-Projects) — Structured collection of semester-wise coursework and systems projects for my BS in Artificial Intelligence at NFC IET Multan. Includes data structures, algorithms, IoT simulators, search engine implementations, and AI academic tools.

OPEN-SOURCE CONTRIBUTIONS

- **drt-hub/drt** — Triage Collaborator on a Python Reverse ETL engine. Designed the dynamic connector registry, added connectors and config validation, and shipped observability support.
- **apache/hugegraph-ai** — Upgraded the Python client CI from 1.3.0 to 1.7.0 and migrated from manual Docker runs to GitHub service containers with health checks.
- **linuxfoundation/crowd.dev** — Audited and corrected outdated technical documentation, Node version, and repository URLs across the project.

EDUCATION

BS Artificial Intelligence
NFC Institute of Engineering & Technology, Multan

BS Mathematics
Virtual University of Pakistan

CERTIFICATIONS & COMPETITIVE PROGRAMMING

- Applied Machine Learning — Verified
- Python AI Developer — Mimo
- Google Cloud Specialization — NAVTTC (ongoing)
- 400+ problems solved on LeetCode, focused on DSA & algorithm optimization.